Cream Separator



Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Cream separator; Mini Cream Separator

What it is used for

A cream separator separates cream from milk by using centrifugal force to divide the lower-density fat-rich portion from the skim milk. Correct assembly, hygienic handling, balanced operation, and timely cleaning help maintain separation efficiency and milk quality.

Main parts

- Feed bowl or inlet receives the milk.
- Rotating bowl contains the separation assembly.
- Disc stack increases the effective separation area where fitted.
- Cream outlet discharges the separated cream.
- Skim milk outlet discharges the lower-fat milk.
- Drive motor or hand-operated mechanism rotates the bowl.
- Frame and base support the separator.
- Flow control regulates milk feed where fitted.
- Cover and locking mechanism enclose the rotating assembly.
- Collection containers receive cream and skim milk.

Before you start

1. Only trained personnel should operate the cream separator.
2. Read the manual and confirm that the separator is suitable for the milk volume and intended use.
3. Inspect the bowl, disc stack, shaft, drive, cover, locking mechanism, outlets, and collection containers for damage.
4. Confirm that all food-contact parts are clean, sanitised, and assembled according to the manufacturer's instructions.
5. Check that the rotating bowl is correctly seated and all required locking components are secured.
6. Place the separator on a stable, level surface and verify the power supply or hand drive is ready.
7. Prepare fresh milk according to the process instructions and use clean, labelled collection containers.

How to use

1. Assemble the separator and install the bowl and disc stack according to the manual.
2. Secure the cover and all locking components before operating the drive.
3. Place clean containers beneath the cream and skim milk outlets.
4. Start the drive and allow the bowl to reach its specified operating speed before feeding milk.

5. Feed milk at the recommended temperature and rate for the model and process.
6. Observe the cream and skim milk outlets and adjust feed or settings only as permitted by the manual.
7. Collect the separated products in their designated containers and prevent cross-contamination.
8. Stop feeding milk before stopping the drive, following the manufacturer's shutdown sequence.
9. Wait until the bowl has stopped completely before opening the cover or removing components.
10. Disconnect power where applicable, disassemble the product-contact parts, and remove milk residues promptly.
11. Clean and sanitise the bowl, disc stack, outlets, and containers according to the approved procedure.
12. Record production quantities and any separation or equipment problems according to facility practice.

⚠ Safety

- Never open the cover or reach into the separator while the bowl is rotating.
- Do not operate with a damaged bowl, loose disc stack, faulty locking mechanism, or excessive vibration.
- Keep hands, hair, clothing, and tools away from moving parts and drive components.
- Disconnect electrical power before disassembly, cleaning, or maintenance of powered models.
- Use approved food-contact cleaning chemicals and follow their dilution and rinsing instructions.
- Keep milk within required time and temperature controls to limit microbial growth.

Care & storage

- Clean and sanitise all milk-contact components after each use.
- Inspect the bowl, disc stack, shaft, seals, and outlets for wear or residue buildup.
- Lubricate only designated points using food-compatible lubricant specified by the manufacturer.
- Keep the separator dry and store removable components in a clean, protected location.
- Arrange authorised service if vibration, leakage, unusual noise, or separation quality problems persist.

Common mistakes to avoid

- Operating before the bowl and disc stack are correctly assembled and locked.
- Feeding milk too quickly or outside the recommended temperature range.
- Opening the cover before the bowl has completely stopped.
- Leaving milk residue in the separator or using unclean collection containers.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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